



# Loan Default Prediction

BY MICHAEL JUDSON

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# Problem definition

## CONTEXT

- Banks offer home loans to obtain profits
- Loans are borrowed by bank customers
- Banks are rigorous while approving loans

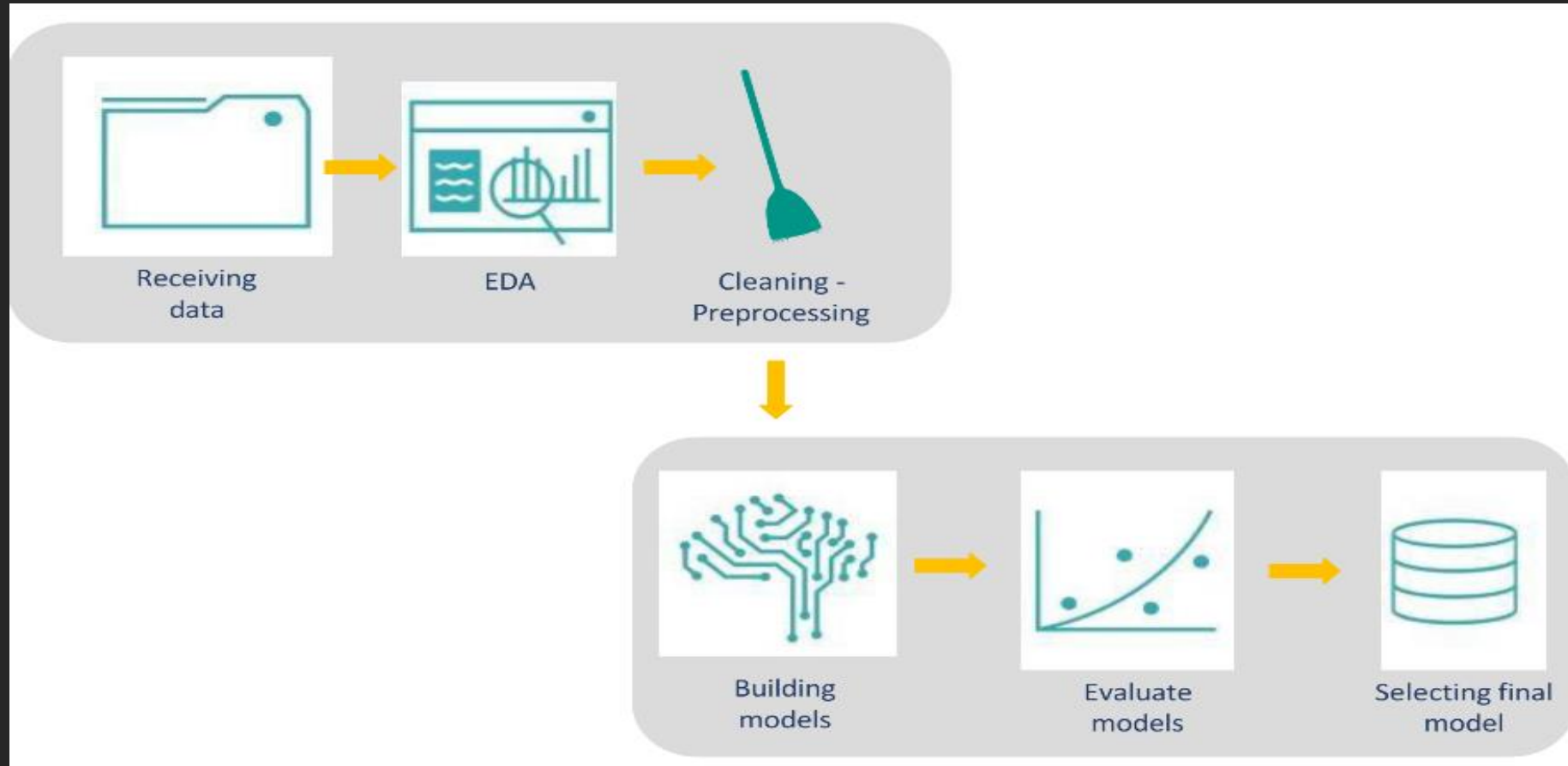
## PROBLEM

- Banks need an effective approval process
- This process is effort-intensive and sensitive to human error and biases

## OBJECTIVES

- Predicts clients who are likely to default on their loan
  - Avoid the risk of misclassification of default loans predicted as non-default loans
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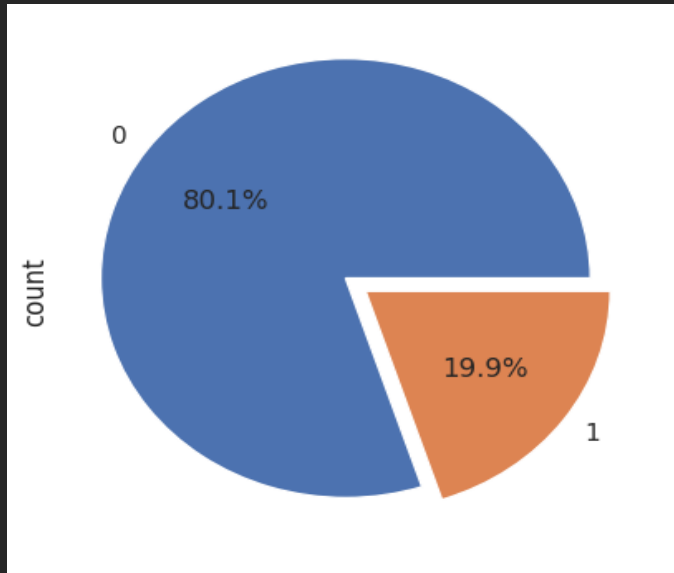
# Solution Approach



\*Image taken from Databricks.com

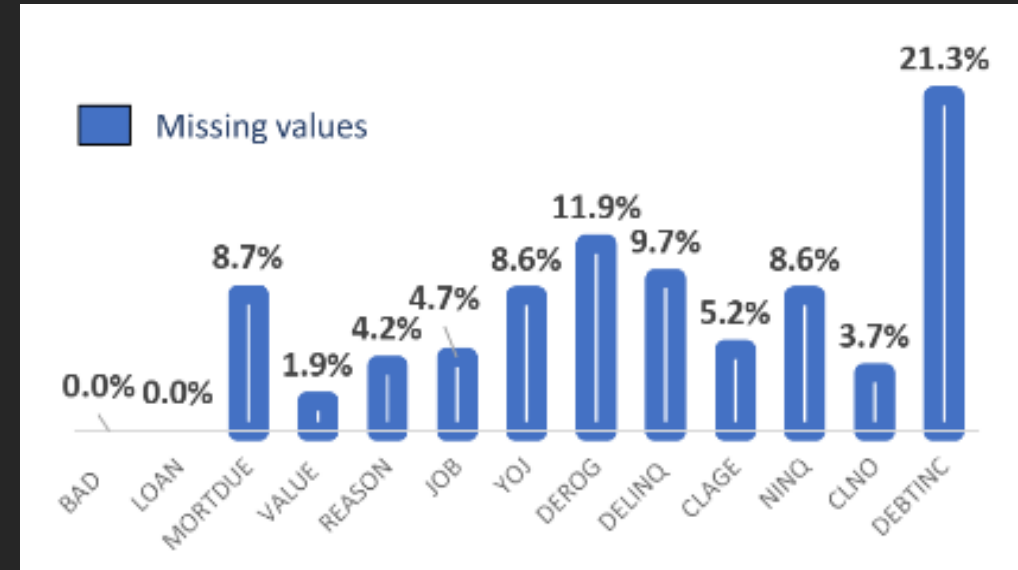
# Data Set Overview

## LOAN DEFAULT RATES



- The dataset contains 5,960 registers and 13 features
- The overall loan default rate is about 20%
- The data is unbalanced in an 80%/20% proportion

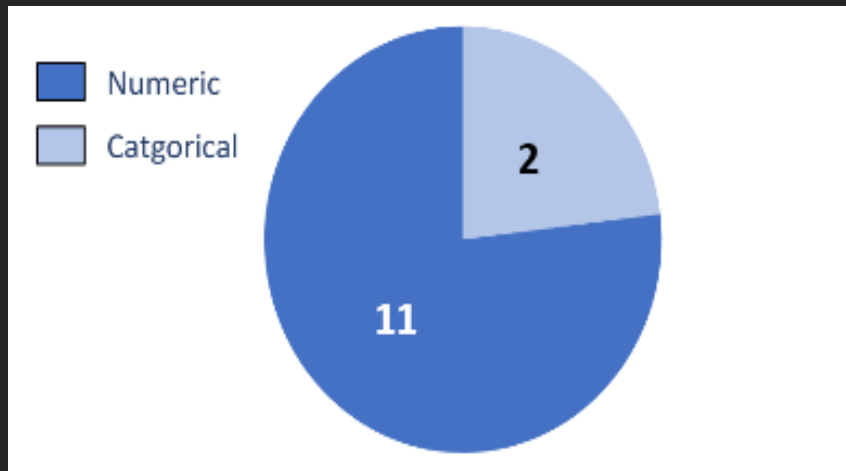
## MISSING VALUES



- All features have missing values except for LOAN and BAD
- Use Median and Mode to fill in missing values

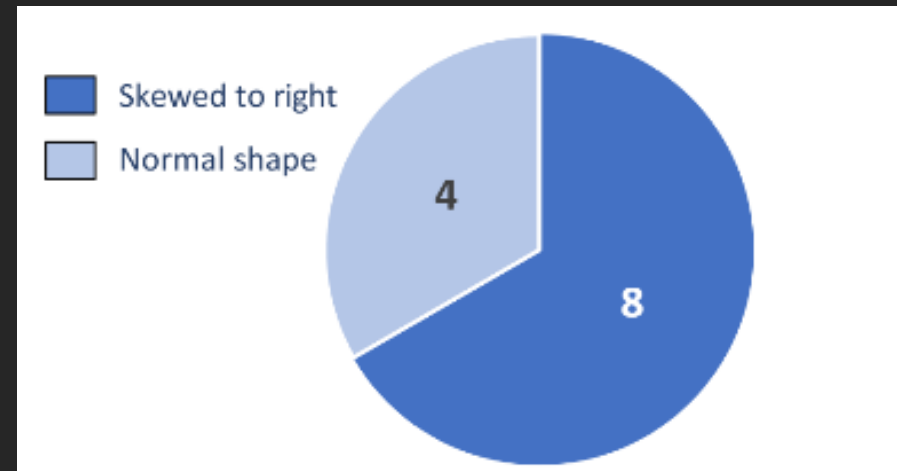
# Data Set Overview

## DATA TYPES



- Data types are distributed with 11 numerical and 2 categorical features
- Categorical data will need to be converted to numerical

## OUTLIERS



- All features have a large amount of outliers
- 8 features are skewed to the right and 4 are normal shape
- Outliers need to be treated for logistic regression

# Proposed model solutions



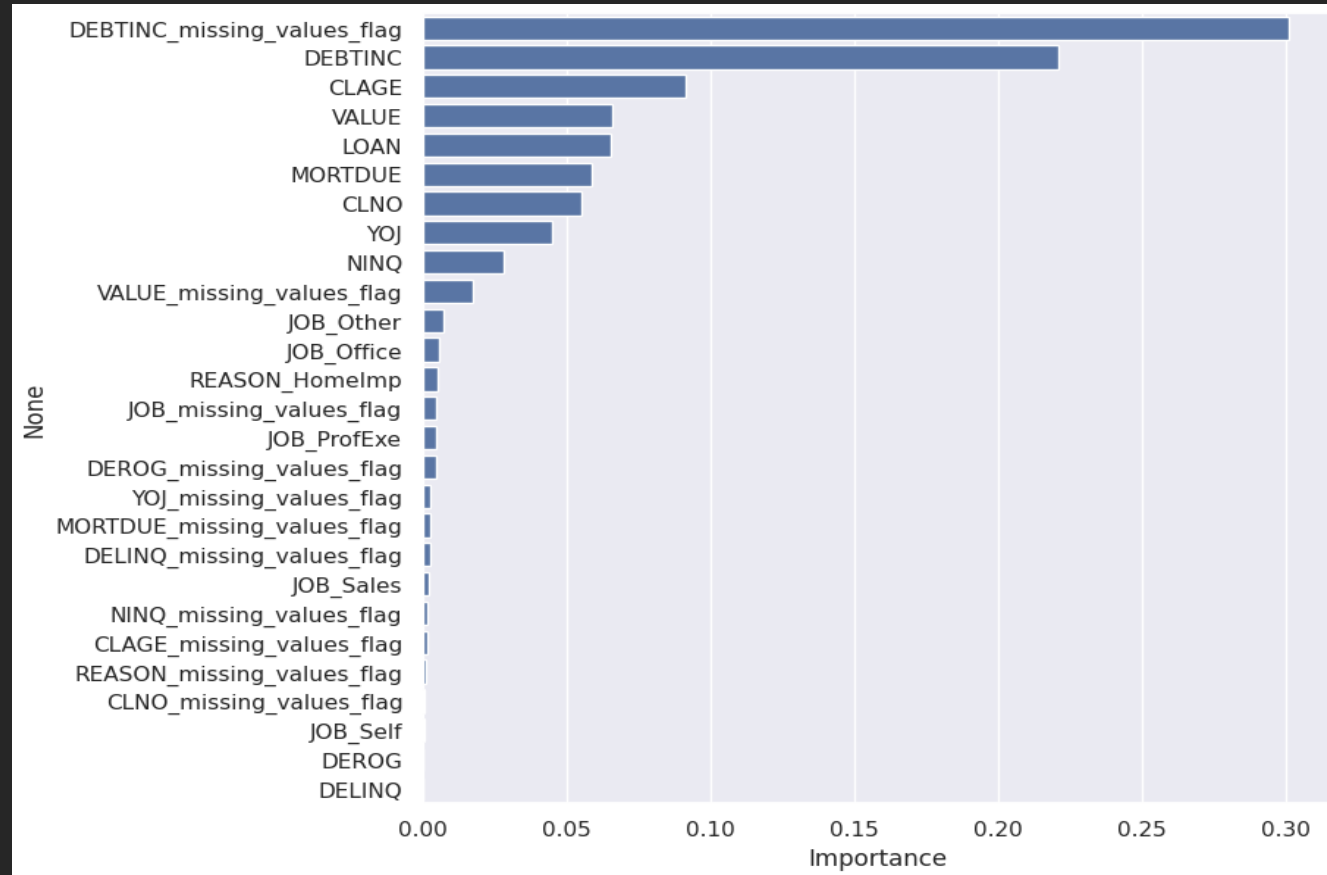
# Comparison of techniques and performances

|                                                | Precision | Recall   | Accuracy |
|------------------------------------------------|-----------|----------|----------|
| Logistic regresstion model                     | 0.595845  | 0.644079 | 0.623043 |
| Logistic regresstion model using 0.6 treshold  | 0.595845  | 0.644079 | 0.623043 |
| Logistic regresstion model using 0.45 treshold | 0.595845  | 0.644079 | 0.623043 |
| Decision Tree classifier                       | 0.801091  | 0.759887 | 0.862975 |
| Tuned Decision Tree classifier                 | 0.749346  | 0.810336 | 0.828300 |
| Random Forest classifier                       | 0.860074  | 0.797783 | 0.893177 |
| Balanced Random Forest classifier              | 0.856855  | 0.782714 | 0.888143 |
| Tuned Random Forest classifier                 | 0.835038  | 0.836363 | 0.891499 |
| Adaboost classifier                            | 0.838270  | 0.768783 | 0.878635 |
| gbc classifier                                 | 0.852236  | 0.783989 | 0.887025 |
| XGBoost classifier                             | 0.868268  | 0.812921 | 0.899888 |
| KNN                                            | 0.786685  | 0.572592 | 0.814318 |
| Tuned KNN                                      | 0.681039  | 0.605317 | 0.798658 |

The Tuned Random Forest classifier is the best model for this problem

- 0.83 recall, 0.89 accuracy, and 0.83 precision

# Model features of importance



DEBTINC values are high showing a positive contribution to identify defaulter

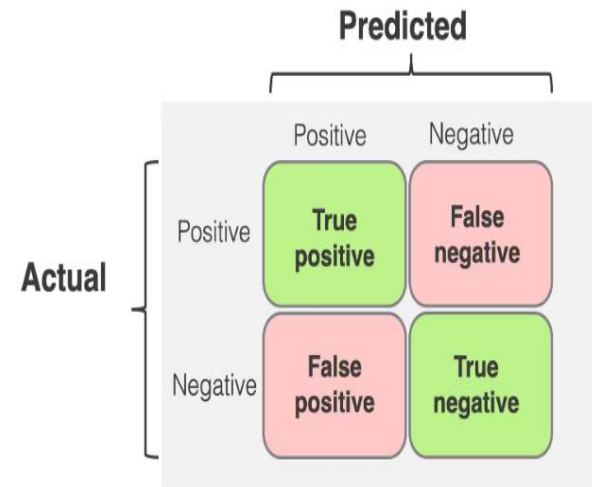
# Proposal for the Final Solution Design

**Tuned Random Forest classifier**

| Precision | Recall   | Accuracy |
|-----------|----------|----------|
| 0.835038  | 0.836363 | 0.891499 |

The **Tuned Random Forest classifier** is the best model for this problem 0.83 recall, 0.89 accuracy, and 0.83 precision

- Recall: Of all the true bad loans, this model identified 83% of them.
- Precision: When loans are predicted to be bad, they are bad 83% of the time.



# Executive summary

- The Tuned Random Forest Classifier can predict loan defaulters 83% of the time they ask for a home loan.
  - This model has the highest recall, 83% for class 1 on the test data, 83% precision score, and 89% accuracy.
  - The most important features to make a proper prediction are DEBINC, DELINQ, and CLAGE
  - Debt-to-Income ratio is the most important feature but has the most missing data (21.3%) which is similar to the proportion of defaulted customers(19.9%)
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# Recommendations and next steps

- Check if there is a way to complete the missing values in the data.
  - Check the possibility to create an alternative business process to manage and take decisions on those clients with no Debt-to-Income ratio available.
  - Create a pilot test with the new model and compare the results with the current manual process before completing the transition to the new model.
  - Explore other machine learning techniques such as feature engineering, dropping columns, support vector machine, and neural networks.
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# Risks and Challenges

## Trust and Transparency

- Risk- Lack of transparency in decision making.
- Challenge- Clear communication, transparency in explaining model functioning.

## Bias in Data

- Risk- Inherent biases in historical data.
- Challenge- Regular monitoring and audits to identify and address biases.

## Model Overfitting

- Risk- Overfitting to training data.
- Challenge- Regular evaluation on diverse datasets, cross validation to prevent overfitting.

## Data Privacy and security

- Risk- Privacy and security concerns with sensitive customer data.
  - Challenge- Compliance with data protection regulations, robust data security measures.
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# Why is recall important to this project?

The model can make two types of wrong predictions:

1. Predicting a client will pay their loan when the client cannot pay.
2. Predicting a client will not pay their loan when the client can pay.

Which is more important?

- **Predicting the client will pay but the client cannot.** I.e., losing money immediately. This would be considered the most important case of wrong predictions because it would have a negative effect on the banks "bottom line" and the bank would not make a profit.

How to reduce this loss?

- The bank would want the RECALL to be maximized. The greater the recall, the higher the chances of minimizing the false negatives or identifying the true positives. This would help increase the bank profit that comes from interests in the form of home loans.
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# Thank You!

